

SOMRUP MUKHERJEE

 Somrup Mukherjee |  somrupmukherjee22@gmail.com |  (+91)9330626107 |  Kolkata, India

PERSONAL PROFILE

Diligent and inquisitive undergraduate student with a good academic record, currently pursuing B.Tech in Optics & Optoelectronics at the Department of Applied Optics & Photonics, University of Calcutta. My passion is centered on Planetary Science, Exoplanets, Planet & Star Formation, Star-Planet Interaction, Optics, Astronomical Instrumentation and related areas.

EDUCATION

University of Calcutta

August 2024 - present

B.Tech in Optics & Optoelectronics Engineering, CGPA: 9.23/10.0 (3rd Semester)

Uttarpara Model School(H.S.)-Class XII

May 2024

West Bengal Council of Higher Secondary Education, Percentage: 85.6%

Uttarpara Model School(H.S.)-Class X

June 2022

West Bengal Board of Secondary Education, Percentage: 94.14%

RESEARCH PROJECTS

● Detection and Characterization of the Exoplanet HD 209458 b

July'26 - Aug'26

[Project Report](#)

- Developed an independent exoplanet detection pipeline using publicly available TESS photometry, including data retrieval, preprocessing, detrending, and transit-signal extraction.
- Applied the Box Least Squares (BLS) method to identify the periodic transit signal and used a Mandel–Agol transit model to fit the phase-folded light curve and characterize the planetary system.
- Recovered the orbital period, transit depth, planet-to-star radius ratio, scaled orbital distance, and orbital inclination, obtaining values within approximately 1–2% of independently published literature results.

● Hot Jupiter Radius Inflation Analysis (Independent Project)

June 2026

[Project Summary](#)

- Analyzed observational exoplanet datasets using Python to study hot Jupiter radius inflation as a function of equilibrium temperature and stellar irradiation.
- Performed data cleaning, preprocessing, correlation studies, and threshold analysis to identify key inflation trends.
- Implemented and compared multiple linear and nonlinear models through statistical fitting, visualization, and physical interpretation.

● Fabry–Perot Interferometer

Jan'26 – Feb'26

- Assembled and aligned a Fabry–Perot interferometer using a laser source and CCD camera to digitally observe interference fringes.
- Gained hands-on experience in optical alignment and basic fringe pattern analysis using computer-based imaging.

EXPERIENCE

Research Intern (Remote)

School of Earth and Space Exploration, Arizona State University

August 2026 – Present

Supervisor:- Dr. Sagnick Mukherjee

- Working on exoplanet atmospheres.

Research Intern (On-Site)

CSIR–Central Scientific Instruments Organisation (CSIO)

July 2026 – August 2026

Supervisor:- Dr. Neha Khatri

- Zemax-Based Optical Design and Simulation of an Off-Axis Solar-Pumped Laser System.
- Designed and simulated off-axis optical configurations in Zemax to optimize solar beam collection and improve coupling efficiency into the laser gain medium.
- Performed ray-tracing and optical aberration analysis to evaluate and enhance focusing accuracy and overall system performance.

TECHNICAL & COMMUNICATION SKILLS

Softwares : MATLAB , ZEMAX.

Programming : Python, C/C++.

Soft Skills : Problem solving, Self-Learning, Adaptability.

Languages : English, Bengali, Hindi.

SELECTED COURSEWORK

Classical Mechanics, Quantum Mechanics (Introductory), Electromagnetism, Foundation of Optical Science, Fundamentals of Optoelectronics, Network Theory & Circuit Analysis, Digital & Analog Electronics & Communication, Engineering Mathematics(I,II & III), Image Science, LASER Physics, Radiometry & Photometry, Signals & Systems.

AWARDS & ACHIEVEMENTS

Recipient of G.P.Birla Scholarship

November 2024 - Present

Secured 1st rank in quiz competition on Astronomy, University of Calcutta

May 2025

ACTIVITIES

Secretary

April 2026 - Present

OPTICA (OSA): University of Calcutta Student Chapter.

- Demonstration of various optical models and theories in different schools and colleges.
- Organize seminars and events.